

A research vessel employs a hybrid propulsion system that includes a pressurized water discharge network composed of two connected pipe sections and a conventional hull structure. The ship must overcome hydrodynamic resistance from both the internal pipe flow and the external hull friction.

The pipe system is composed of:

- **Section A:** length $L_A = 30$ m, diameter $D_A = 0.3$ m, roughness $k_A = 0.3$ mm
- **Section B:** length $L_B = 20$ m, diameter $D_B = 0.2$ m, roughness $k_B = 1.2$ mm

The total flow rate through the system is $Q = 0.08$ m³/s. Seawater is assumed with density $\rho = 1025$ kg/m³ and kinematic viscosity $\nu = 1.1 \times 10^{-6}$ m²/s.

The ship's hull has length $L = 85$ m, draft $T = 6$ m, and an estimated surface roughness $k_s = 4$ mm. The vessel moves at a constant velocity $V = 9$ m/s and the propeller has an efficiency of $\eta_S = 0.78$.

Tasks

- Calculate the **mean velocity** of the flow in each pipe section.
- Compute the **Reynolds number** in each section. Indicate whether the flow is laminar or turbulent.
- Estimate the **total pressure loss** in both pipe sections using the Darcy-Weisbach equation:

$$\Delta p_V = \lambda \cdot \frac{l}{d} \cdot \frac{\rho}{2} \cdot \bar{u}^2$$

Use the following to determine the friction factor λ :

- For laminar flow: $\lambda = \frac{64}{Re}$
- For fully rough turbulent flow:

$$\frac{1}{\sqrt{\lambda}} = 1.74 - 2 \log_{10} \left(\frac{2k_s}{d} \right)$$

- Calculate the **friction coefficient for the hull** using the Prandtl-Schlichting formula:

$$c_f = \left(1.89 + 1.62 \cdot \log_{10} \left(\frac{L}{k_s} \right) \right)^{-2.5}$$

Then determine the **total frictional force** on both sides of the hull.

- Determine the **total effective power required** to overcome:
 - The pressure losses in the pipe network
 - The hull friction resistance

Then calculate the **required engine power output** considering the propeller efficiency η_S .